Figure 5.9 shows where this black hole is on the night sky in summer. It is to the lower right of the constellation Sagittarius, the teapot, at the × labeled “Galactic Center.”

A massive black hole inhabits the core of nearly every big galaxy in our universe. Many of these are as heavy as Gargantua (100 million Suns), or even heavier. The heaviest yet measured is 17 billion times more massive than the Sun; it resides at the center of a galaxy whose name is NGC1277, 250 million light-years from Earth—roughly a tenth of the way to the edge of the visible universe.

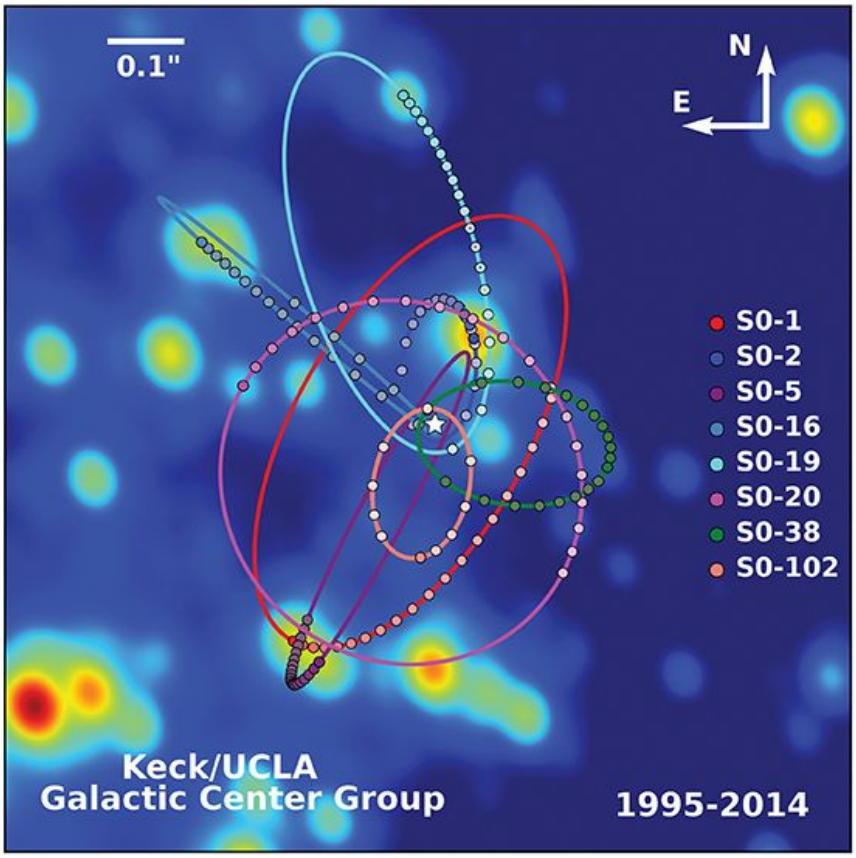


Fig. 5.8. Observed orbits of stars around the massive black hole at the center of our Milky Way galaxy, as measured by Andrea Ghez and colleagues.